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and the footprint are common and dangerous. Even a single mismatch can make the entire IC unusable on the PCB.

Keep the layer sequence the same. All footprint information is distributed across multiple layers, such as copper, silkscreen, courtyard, component centre, and assembly information. I maintain a fixed layer sequence across all components. If one component uses layer five for silkscreen, every component must follow the same rule. Inconsistent layer usage creates major documentation and assembly issues.

Create your own footprint. I strongly believe in creating my own footprints. Many engineers hesitate and prefer downloading footprints from random websites or vendors. That is risky. Someone else could have made a mistake, and I would be blindly trusting it.

Even when footprints come from official vendors, their layer conventions may not match mine.

Directly importing them can corrupt my library and increase layer count unnecessarily. If I reuse vendor footprints, I only copy the copper layers first, then manually place courtyard and silkscreen information into my own defined layers. This keeps the library clean and consistent.

The datasheet is your best friend. I learned this the hard way. I once used an NXP IC where a website listed the package as SOIC 16. Based on that, I assigned an SOIC 16 footprint. When the PCB reached assembly, the IC did not fit the pads. Only after checking the datasheet did I realise the package was actually SO 16, not SOIC 16. The difference was just two letters, but physically the SO package is almost half the size. The IC ended up sitting on one side of the pads, and the rest had to be wired manually.

This mistake happened because I trusted the website information instead of the datasheet. The website is still incorrect even today.

The lesson is simple. Never trust secondary sources. Always refer to the vendor datasheet. There are no shortcuts here.

From schematic to PCB layout

Coming to the schematic part, the first question I simply ask is, "What exactly do we need to do in the schematic?" The answer is:

Capture your schematics clearly.

I cannot stress this enough. A schematic is not just about picking components and connecting pins until the ERC passes. When someone looks at my schematic, they should immediately understand what is happening. This is where the signal comes from. This is where it goes. These are the interfaces. A schematic must communicate intent. If it does not, it has failed its purpose.

Pay attention to the pin assignment. Pin assignment mistakes are extremely common. I have seen many engineers make errors with SPI communication.